

5_Python_Project_Extracting_Stock_Data_Using_a_Python_Library_2

January 30, 2024

APPLIED LINGUISTICS GRADUATE PROGRAMME (LAEL)

1 Extracting Stock Data Using a Python Library

```
[1]: import sys
!{sys.executable} -m pip install yfinance==0.2.4
```

```
Defaulting to user installation because normal site-packages is not writeable
Looking in links: /usr/share/pip-wheels
Requirement already satisfied: yfinance==0.2.4 in
/home/a9324a50-a59d-4d3b-9871-edda0af6600a/.local/lib/python3.11/site-packages
(0.2.4)
Requirement already satisfied: pandas>=1.3.0 in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from yfinance==0.2.4) (1.5.3)
Requirement already satisfied: numpy>=1.16.5 in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from yfinance==0.2.4) (1.24.3)
Requirement already satisfied: requests>=2.26 in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from yfinance==0.2.4) (2.29.0)
Requirement already satisfied: multitasking>=0.0.7 in
/home/a9324a50-a59d-4d3b-9871-edda0af6600a/.local/lib/python3.11/site-packages
(from yfinance==0.2.4) (0.0.11)
Requirement already satisfied: lxml>=4.9.1 in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from yfinance==0.2.4) (4.9.1)
Requirement already satisfied: appdirs>=1.4.4 in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from yfinance==0.2.4) (1.4.4)
Requirement already satisfied: pytz>=2022.5 in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from yfinance==0.2.4) (2022.7)
Requirement already satisfied: frozendict>=2.3.4 in
/home/a9324a50-a59d-4d3b-9871-edda0af6600a/.local/lib/python3.11/site-packages
(from yfinance==0.2.4) (2.3.8)
Requirement already satisfied: cryptography>=3.3.2 in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from yfinance==0.2.4) (39.0.1)
Requirement already satisfied: beautifulsoup4>=4.11.1 in
/opt/conda/envs/anaconda-panel-2023.05-py310/lib/python3.11/site-packages (from
yfinance==0.2.4) (4.12.2)
Requirement already satisfied: html5lib>=1.1 in
```

```

/home/a9324a50-a59d-4d3b-9871-edda0af6600a/.local/lib/python3.11/site-packages
(from yfinance==0.2.4) (1.1)
Requirement already satisfied: soupsieve>1.2 in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from
beautifulsoup4>=4.11.1->yfinance==0.2.4) (2.4)
Requirement already satisfied: cffi>=1.12 in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from
cryptography>=3.3.2->yfinance==0.2.4) (1.15.1)
Requirement already satisfied: six>=1.9 in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from
html5lib>=1.1->yfinance==0.2.4) (1.16.0)
Requirement already satisfied: webencodings in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from
html5lib>=1.1->yfinance==0.2.4) (0.5.1)
Requirement already satisfied: python-dateutil>=2.8.1 in
/opt/conda/envs/anaconda-panel-2023.05-py310/lib/python3.11/site-packages (from
pandas>=1.3.0->yfinance==0.2.4) (2.8.2)
Requirement already satisfied: charset-normalizer<4,>=2 in
/opt/conda/envs/anaconda-panel-2023.05-py310/lib/python3.11/site-packages (from
requests>=2.26->yfinance==0.2.4) (2.0.4)
Requirement already satisfied: idna<4,>=2.5 in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from
requests>=2.26->yfinance==0.2.4) (3.4)
Requirement already satisfied: urllib3<1.27,>=1.21.1 in
/opt/conda/envs/anaconda-panel-2023.05-py310/lib/python3.11/site-packages (from
requests>=2.26->yfinance==0.2.4) (1.26.16)
Requirement already satisfied: certifi>=2017.4.17 in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from
requests>=2.26->yfinance==0.2.4) (2023.5.7)
Requirement already satisfied: pycparser in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from
cffi>=1.12->cryptography>=3.3.2->yfinance==0.2.4) (2.21)

```

```

[2]: import yfinance as yf
      import pandas as pd

```

1.1 Apple

```

[3]: apple = yf.Ticker('AAPL')

```

```

[4]: # Make sure the file 'apple.json' is in the directory
      import json
      with open('apple.json') as json_file:
          apple_info = json.load(json_file)
      print(apple_info)

```

```

{'zip': '95014', 'sector': 'Technology', 'fullTimeEmployees': 100000,

```

'longBusinessSummary': 'Apple Inc. designs, manufactures, and markets smartphones, personal computers, tablets, wearables, and accessories worldwide. It also sells various related services. In addition, the company offers iPhone, a line of smartphones; Mac, a line of personal computers; iPad, a line of multi-purpose tablets; AirPods Max, an over-ear wireless headphone; and wearables, home, and accessories comprising AirPods, Apple TV, Apple Watch, Beats products, HomePod, and iPod touch. Further, it provides AppleCare support services; cloud services store services; and operates various platforms, including the App Store that allow customers to discover and download applications and digital content, such as books, music, video, games, and podcasts. Additionally, the company offers various services, such as Apple Arcade, a game subscription service; Apple Music, which offers users a curated listening experience with on-demand radio stations; Apple News+, a subscription news and magazine service; Apple TV+, which offers exclusive original content; Apple Card, a co-branded credit card; and Apple Pay, a cashless payment service, as well as licenses its intellectual property. The company serves consumers, and small and mid-sized businesses; and the education, enterprise, and government markets. It distributes third-party applications for its products through the App Store. The company also sells its products through its retail and online stores, and direct sales force; and third-party cellular network carriers, wholesalers, retailers, and resellers. Apple Inc. was incorporated in 1977 and is headquartered in Cupertino, California.', 'city': 'Cupertino', 'phone': '408 996 1010', 'state': 'CA', 'country': 'United States', 'companyOfficers': [], 'website': 'https://www.apple.com', 'maxAge': 1, 'address1': 'One Apple Park Way', 'industry': 'Consumer Electronics', 'ebitdaMargins': 0.33890998, 'profitMargins': 0.26579002, 'grossMargins': 0.43019, 'operatingCashflow': 112241000448, 'revenueGrowth': 0.112, 'operatingMargins': 0.309, 'ebitda': 128217997312, 'targetLowPrice': 160, 'recommendationKey': 'buy', 'grossProfits': 152836000000, 'freeCashflow': 80153247744, 'targetMedianPrice': 199.5, 'currentPrice': 177.77, 'earningsGrowth': 0.25, 'currentRatio': 1.038, 'returnOnAssets': 0.19875, 'numberOfAnalystOpinions': 44, 'targetMeanPrice': 193.53, 'debtToEquity': 170.714, 'returnOnEquity': 1.45567, 'targetHighPrice': 215, 'totalCash': 63913000960, 'totalDebt': 122797998080, 'totalRevenue': 378323009536, 'totalCashPerShare': 3.916, 'financialCurrency': 'USD', 'revenuePerShare': 22.838, 'quickRatio': 0.875, 'recommendationMean': 1.8, 'exchange': 'NMS', 'shortName': 'Apple Inc.', 'longName': 'Apple Inc.', 'exchangeTimezoneName': 'America/New_York', 'exchangeTimezoneShortName': 'EDT', 'isEsgPopulated': False, 'gmtOffSetMilliseconds': '-14400000', 'quoteType': 'EQUITY', 'symbol': 'AAPL', 'messageBoardId': 'finmb_24937', 'market': 'us_market', 'annualHoldingsTurnover': None, 'enterpriseToRevenue': 7.824, 'beta3Year': None, 'enterpriseToEbitda': 23.086, '52WeekChange': 0.4549594, 'morningStarRiskRating': None, 'forwardEps': 6.56, 'revenueQuarterlyGrowth': None, 'sharesOutstanding': 16319399936, 'fundInceptionDate': None, 'annualReportExpenseRatio': None, 'totalAssets': None, 'bookValue': 4.402, 'sharesShort': 111286790, 'sharesPercentSharesOut': 0.0068, 'fundFamily': None, 'lastFiscalYearEnd': 1632528000, 'heldPercentInstitutions': 0.59397, 'netIncomeToCommon': 100554997760, 'trailingEps': 6.015, 'lastDividendValue': 0.22, 'SandP52WeekChange': 0.15217662, 'priceToBook': 40.38392,

```
'heldPercentInsiders': 0.0007, 'nextFiscalYearEnd': 1695600000, 'yield': None,
'mostRecentQuarter': 1640390400, 'shortRatio': 1.21,
'sharesShortPreviousMonthDate': 1644883200, 'floatShares': 16302795170, 'beta':
1.185531, 'enterpriseValue': 2959991898112, 'priceHint': 2,
'threeYearAverageReturn': None, 'lastSplitDate': 1598832000, 'lastSplitFactor':
'4:1', 'legalType': None, 'lastDividendDate': 1643932800,
'morningStarOverallRating': None, 'earningsQuarterlyGrowth': 0.204,
'priceToSalesTrailing12Months': 7.668314, 'dateShortInterest': 1647302400,
'pegRatio': 1.94, 'ytdReturn': None, 'forwardPE': 27.099087, 'lastCapGain':
None, 'shortPercentOfFloat': 0.0068, 'sharesShortPriorMonth': 108944701,
'impliedSharesOutstanding': 0, 'category': None, 'fiveYearAverageReturn': None,
'previousClose': 178.96, 'regularMarketOpen': 178.55, 'twoHundredDayAverage':
156.03505, 'trailingAnnualDividendYield': 0.004833482, 'payoutRatio': 0.1434,
'volume24Hr': None, 'regularMarketDayHigh': 179.61, 'navPrice': None,
'averageDailyVolume10Day': 93823630, 'regularMarketPreviousClose': 178.96,
'fiftyDayAverage': 166.498, 'trailingAnnualDividendRate': 0.865, 'open': 178.55,
'toCurrency': None, 'averageVolume10days': 93823630, 'expireDate': None,
'algorithm': None, 'dividendRate': 0.88, 'exDividendDate': 1643932800,
'circulatingSupply': None, 'startDate': None, 'regularMarketDayLow': 176.7,
'currency': 'USD', 'trailingPE': 29.55445, 'regularMarketVolume': 92633154,
'lastMarket': None, 'maxSupply': None, 'openInterest': None, 'marketCap':
2901099675648, 'volumeAllCurrencies': None, 'strikePrice': None,
'averageVolume': 95342043, 'dayLow': 176.7, 'ask': 178.53, 'askSize': 800,
'volume': 92633154, 'fiftyTwoWeekHigh': 182.94, 'fromCurrency': None,
'fiveYearAvgDividendYield': 1.13, 'fiftyTwoWeekLow': 122.25, 'bid': 178.4,
'tradeable': False, 'dividendYield': 0.005, 'bidSize': 3200, 'dayHigh': 179.61,
'regularMarketPrice': 177.77, 'preMarketPrice': 178.38, 'logo_url':
'https://logo.clearbit.com/apple.com'}
```

```
[5]: print(apple_info['country'])
```

United States

```
[6]: apple_share_price_data = apple.history(period = 'max')
apple_share_price_data.head()
```

```
[6]:
```

	Open	High	Low	Close	Volume \
Date					
1980-12-12 00:00:00-05:00	0.099584	0.100017	0.099584	0.099584	469033600
1980-12-15 00:00:00-05:00	0.094821	0.094821	0.094388	0.094388	175884800
1980-12-16 00:00:00-05:00	0.087894	0.087894	0.087461	0.087461	105728000
1980-12-17 00:00:00-05:00	0.089625	0.090058	0.089625	0.089625	86441600
1980-12-18 00:00:00-05:00	0.092224	0.092657	0.092224	0.092224	73449600

	Dividends	Stock Splits
Date		
1980-12-12 00:00:00-05:00	0.0	0.0

1980-12-15 00:00:00-05:00	0.0	0.0
1980-12-16 00:00:00-05:00	0.0	0.0
1980-12-17 00:00:00-05:00	0.0	0.0
1980-12-18 00:00:00-05:00	0.0	0.0

```
[7]: apple_share_price_data.reset_index(inplace = True)
apple_share_price_data.head()
```

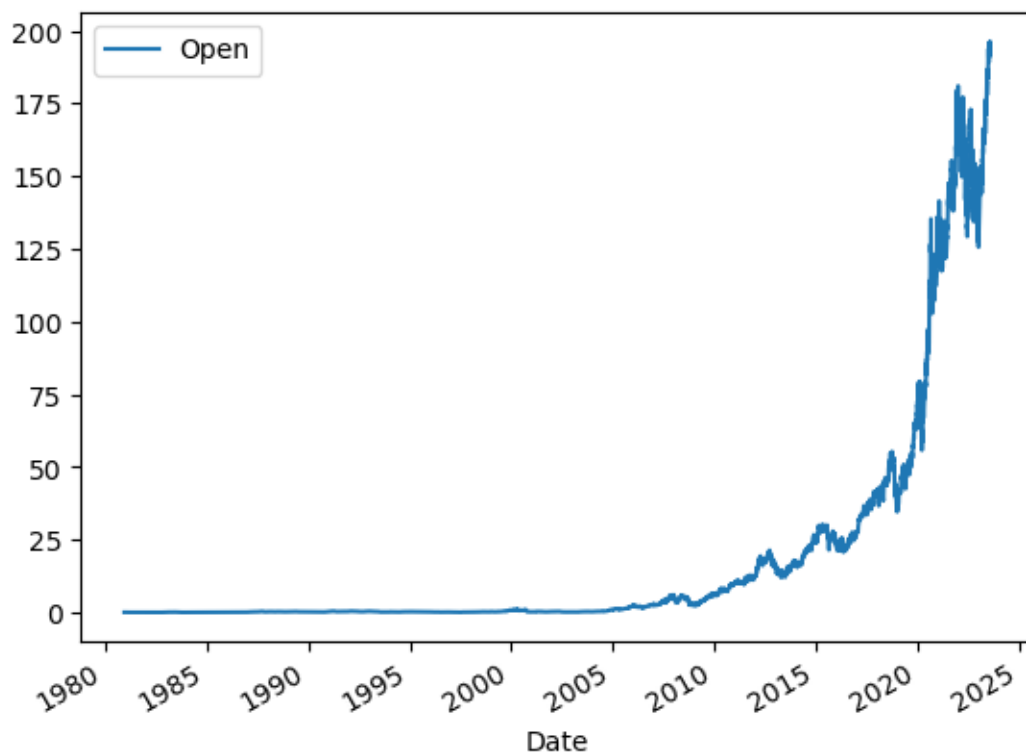
```
[7]:
```

	Date	Open	High	Low	Close	\
0	1980-12-12 00:00:00-05:00	0.099584	0.100017	0.099584	0.099584	
1	1980-12-15 00:00:00-05:00	0.094821	0.094821	0.094388	0.094388	
2	1980-12-16 00:00:00-05:00	0.087894	0.087894	0.087461	0.087461	
3	1980-12-17 00:00:00-05:00	0.089625	0.090058	0.089625	0.089625	
4	1980-12-18 00:00:00-05:00	0.092224	0.092657	0.092224	0.092224	

	Volume	Dividends	Stock Splits
0	469033600	0.0	0.0
1	175884800	0.0	0.0
2	105728000	0.0	0.0
3	86441600	0.0	0.0
4	73449600	0.0	0.0

```
[8]: apple_share_price_data.plot(x = 'Date', y = 'Open')
```

```
[8]: <Axes: xlabel='Date'>
```

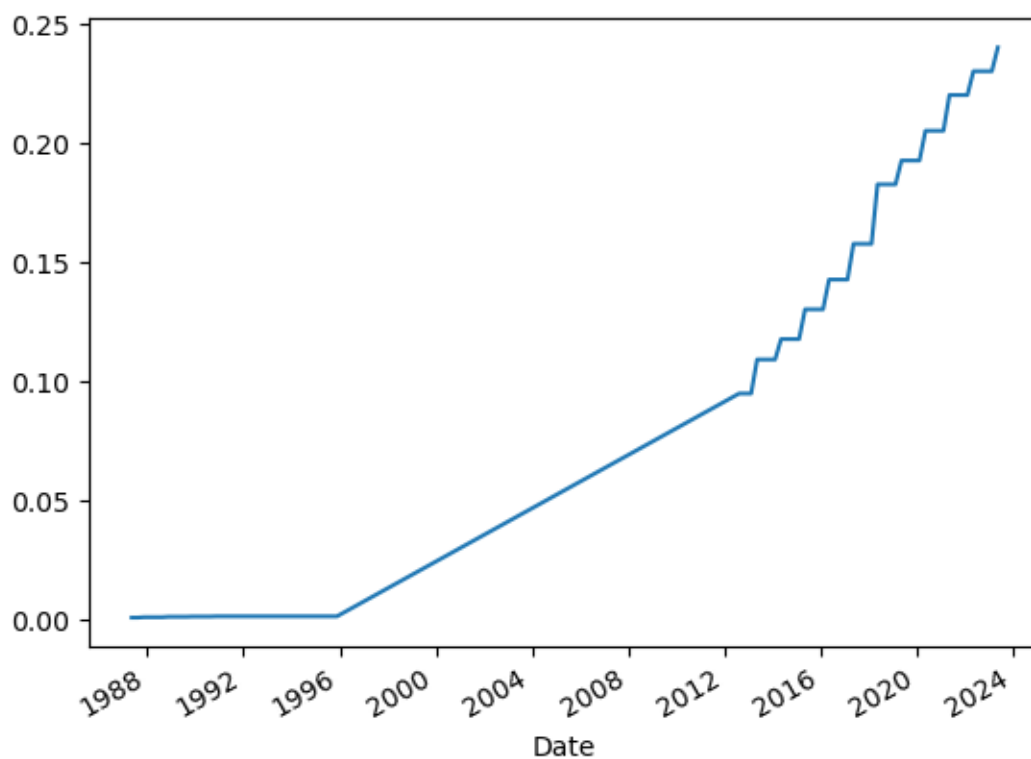


```
[9]: print(type(apple.dividends))  
      print(apple.dividends)
```

```
<class 'pandas.core.series.Series'>  
Date  
1987-05-11 00:00:00-04:00    0.000536  
1987-08-10 00:00:00-04:00    0.000536  
1987-11-17 00:00:00-05:00    0.000714  
1988-02-12 00:00:00-05:00    0.000714  
1988-05-16 00:00:00-04:00    0.000714  
...  
2022-05-06 00:00:00-04:00    0.230000  
2022-08-05 00:00:00-04:00    0.230000  
2022-11-04 00:00:00-04:00    0.230000  
2023-02-10 00:00:00-05:00    0.230000  
2023-05-12 00:00:00-04:00    0.240000  
Name: Dividends, Length: 79, dtype: float64
```

```
[10]: apple.dividends.plot()
```

```
[10]: <Axes: xlabel='Date'>
```



1.2 AMD

```
[11]: amd = yf.Ticker('AMD')
```

```
[12]: # Make sure the file 'amd.json' is in the directory  
import json  
with open('amd.json') as json_file:  
    amd_info = json.load(json_file)  
print(amd_info)
```

```
{'zip': '95054', 'sector': 'Technology', 'fullTimeEmployees': 15500,  
'longBusinessSummary': 'Advanced Micro Devices, Inc. operates as a semiconductor  
company worldwide. The company operates in two segments, Computing and Graphics;  
and Enterprise, Embedded and Semi-Custom. Its products include x86  
microprocessors as an accelerated processing unit, chipsets, discrete and  
integrated graphics processing units (GPUs), data center and professional GPUs,  
and development services; and server and embedded processors, and semi-custom  
System-on-Chip (SoC) products, development services, and technology for game  
consoles. The company provides processors for desktop and notebook personal  
computers under the AMD Ryzen, AMD Ryzen PRO, Ryzen Threadripper, Ryzen  
Threadripper PRO, AMD Athlon, AMD Athlon PRO, AMD FX, AMD A-Series, and AMD PRO  
A-Series processors brands; discrete GPUs for desktop and notebook PCs under the  
AMD Radeon graphics, AMD Embedded Radeon graphics brands; and professional  
graphics products under the AMD Radeon Pro and AMD FirePro graphics brands. It  
also offers Radeon Instinct, Radeon PRO V-series, and AMD Instinct accelerators  
for servers; chipsets under the AMD trademark; microprocessors for servers under  
the AMD EPYC; embedded processor solutions under the AMD Athlon, AMD Geode, AMD  
Ryzen, AMD EPYC, AMD R-Series, and G-Series processors brands; and customer-  
specific solutions based on AMD CPU, GPU, and multi-media technologies, as well  
as semi-custom SoC products. It serves original equipment manufacturers, public  
cloud service providers, original design manufacturers, system integrators,  
independent distributors, online retailers, and add-in-board manufacturers  
through its direct sales force, independent distributors, and sales  
representatives. The company was incorporated in 1969 and is headquartered in  
Santa Clara, California.', 'city': 'Santa Clara', 'phone': '408 749 4000',  
'state': 'CA', 'country': 'United States', 'companyOfficers': [], 'website':  
'https://www.amd.com', 'maxAge': 1, 'address1': '2485 Augustine Drive',  
'industry': 'Semiconductors', 'ebitdaMargins': 0.24674, 'profitMargins':  
0.19240999, 'grossMargins': 0.48248002, 'operatingCashflow': 3520999936,  
'revenueGrowth': 0.488, 'operatingMargins': 0.22198, 'ebitda': 4055000064,  
'targetLowPrice': 107, 'recommendationKey': 'buy', 'grossProfits': 7929000000,  
'freeCashflow': 3122749952, 'targetMedianPrice': 150, 'currentPrice': 119.22,  
'earningsGrowth': -0.454, 'currentRatio': 2.024, 'returnOnAssets': 0.21327,  
'numberOfAnalystOpinions': 38, 'targetMeanPrice': 152.02, 'debtToEquity': 9.764,  
'returnOnEquity': 0.47428, 'targetHighPrice': 200, 'totalCash': 3608000000,  
'totalDebt': 732000000, 'totalRevenue': 16433999872, 'totalCashPerShare': 3.008,
```

```

'financialCurrency': 'USD', 'revenuePerShare': 13.548, 'quickRatio': 1.49,
'recommendationMean': 2.2, 'exchange': 'NMS', 'shortName': 'Advanced Micro
Devices, Inc.', 'longName': 'Advanced Micro Devices, Inc.',
'exchangeTimezoneName': 'America/New_York', 'exchangeTimezoneShortName': 'EDT',
'isEsgPopulated': False, 'gmtOffSetMilliseconds': '-14400000', 'quoteType':
'EQUITY', 'symbol': 'AMD', 'messageBoardId': 'finmb_168864', 'market':
'us_market', 'annualHoldingsTurnover': None, 'enterpriseToRevenue': 8.525,
'beta3Year': None, 'enterpriseToEbitda': 34.551, '52WeekChange': 0.51966953,
'morningStarRiskRating': None, 'forwardEps': 4.72, 'revenueQuarterlyGrowth':
None, 'sharesOutstanding': 1627360000, 'fundInceptionDate': None,
'annualReportExpenseRatio': None, 'totalAssets': None, 'bookValue': 6.211,
'sharesShort': 27776129, 'sharesPercentSharesOut': 0.0171, 'fundFamily': None,
'lastFiscalYearEnd': 1640390400, 'heldPercentInstitutions': 0.52896,
'netIncomeToCommon': 3161999872, 'trailingEps': 2.57, 'lastDividendValue':
0.005, 'SandP52WeekChange': 0.15217662, 'priceToBook': 19.194977,
'heldPercentInsiders': 0.00328, 'nextFiscalYearEnd': 1703462400, 'yield': None,
'mostRecentQuarter': 1640390400, 'shortRatio': 0.24,
'sharesShortPreviousMonthDate': 1644883200, 'floatShares': 1193798619, 'beta':
1.848425, 'enterpriseValue': 140104957952, 'priceHint': 2,
'threeYearAverageReturn': None, 'lastSplitDate': 966902400, 'lastSplitFactor':
'2:1', 'legalType': None, 'lastDividendDate': 798940800,
'morningStarOverallRating': None, 'earningsQuarterlyGrowth': -0.453,
'priceToSalesTrailing12Months': 11.805638, 'dateShortInterest': 1647302400,
'pegRatio': 0.99, 'ytdReturn': None, 'forwardPE': 25.258476, 'lastCapGain':
None, 'shortPercentOfFloat': 0.0171, 'sharesShortPriorMonth': 88709340,
'impliedSharesOutstanding': 0, 'category': None, 'fiveYearAverageReturn': None,
'previousClose': 123.23, 'regularMarketOpen': 123.04, 'twoHundredDayAverage':
116.6998, 'trailingAnnualDividendYield': 0, 'payoutRatio': 0, 'volume24Hr':
None, 'regularMarketDayHigh': 125.66, 'navPrice': None,
'averageDailyVolume10Day': 102167370, 'regularMarketPreviousClose': 123.23,
'fiftyDayAverage': 115.95, 'trailingAnnualDividendRate': 0, 'open': 123.04,
'toCurrency': None, 'averageVolume10days': 102167370, 'expireDate': None,
'algorithm': None, 'dividendRate': None, 'exDividendDate': 798940800,
'circulatingSupply': None, 'startDate': None, 'regularMarketDayLow': 118.59,
'currency': 'USD', 'trailingPE': 46.389107, 'regularMarketVolume': 99476946,
'lastMarket': None, 'maxSupply': None, 'openInterest': None, 'marketCap':
194013855744, 'volumeAllCurrencies': None, 'strikePrice': None, 'averageVolume':
102428813, 'dayLow': 118.59, 'ask': 117.24, 'askSize': 1100, 'volume': 99476946,
'fiftyTwoWeekHigh': 164.46, 'fromCurrency': None, 'fiveYearAvgDividendYield':
None, 'fiftyTwoWeekLow': 72.5, 'bid': 117.24, 'tradeable': False,
'dividendYield': None, 'bidSize': 900, 'dayHigh': 125.66, 'regularMarketPrice':
119.22, 'preMarketPrice': 116.98, 'logo_url':
'https://logo.clearbit.com/amd.com'}

```

```
[13]: print(amd_info['country'])
```

United States


```
[14]: print(amd_info['sector'])
```

Technology

```
[15]: amd_share_price_data = amd.history(period = 'max')
amd_share_price_data.reset_index(inplace = True)
print(amd_share_price_data.iloc[0, amd_share_price_data.columns.
↳get_loc('Volume')])
```

219600

```
[ ]:
```